Full-Custom Design and Analysis of a Two-Stage CMOS Operational Amplifier in Cadence Virtuoso

DC, AC, Transient and Stability (Loop-Gain / Phase-Margin) Analysis with Design Calculations

Technical Design Report

Tool Used: Cadence Virtuoso (Schematic Editor ,ADE-Explorer)

Technology: CMOS , 45nm

1. Objective

The objective of this project is to carry out the full-custom design of a two-stage CMOS operational amplifier (op-amp) at the transistor level using the Cadence Virtuoso schematic editor, and to characterise its performance through DC, AC, transient and stability (loop-gain) simulations run from the Analog Design Environment (ADE-L / ADE Explorer). The report covers the working principle of the two-stage topology, the test-bench configuration used for each analysis, the theoretical design equations governing each performance metric, hand/derived calculations based on the specified bias conditions, and a summary and discussion of the values obtained from ADE Explorer.

2. Introduction to the Two-Stage CMOS Operational Amplifier

Operational Amplifier

An operational amplifier is a high-gain, differential-input, single-ended (or differential) output voltage amplifier that forms the fundamental analog building block of almost every mixed-signal and analog system - filters, ADCs/DACs, references, comparators, regulators and instrumentation front ends. An ideal op-amp exhibits infinite open-loop gain, infinite input impedance, zero output impedance and infinite bandwidth. A practical CMOS op-amp approaches this behaviour within the limits imposed by device physics, supply voltage, bias current and parasitic capacitances.

Why a Two-Stage Topology?

A single differential pair (single-stage / telescopic or folded-cascode amplifier) can provide high gain, but its output swing is restricted because the output node is stacked with multiple current-source/cascode transistors. Many applications - and this one in particular - need both high DC gain and close-to-rail-to-rail output swing while driving a purely capacitive load. The two-stage architecture solves this by separating the two requirements into two cascaded stages:

- Stage 1 (Differential trans-conductance stage): converts the small differential input voltage into a differential/single-ended current and provides the first block of voltage gain along with a high input impedance.
- Stage 2 (Common-source trans-impedance / voltage stage): takes the signal from the first stage and provides additional voltage gain while allowing the output node to swing close to both supply rails, since it is a single transistor loaded by a current source rather than a cascode stack.

The overall open-loop DC gain of the amplifier is the product of the gains of the two stages

$$Av = Av1 \times Av2$$

which is why a two-stage design can achieve 45-60 dB of gain (or more, with cascoding) using only two low-impedance stages, at the cost of requiring frequency compensation because the amplifier now has two dominant internal poles.

Functional Description of the Circuit

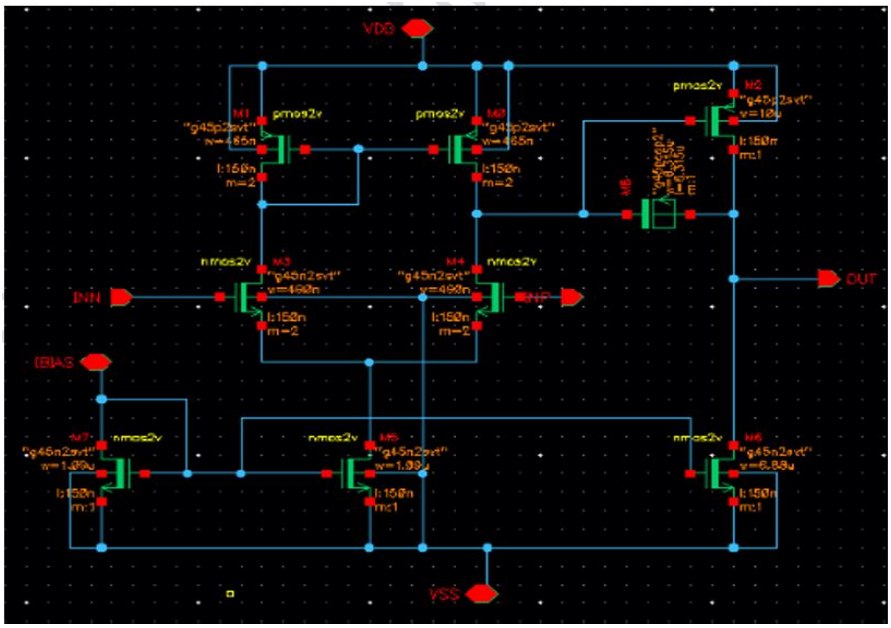
The complete op-amp used in this design consists of the following functional blocks:

- A current-mirror bias generator that converts the external reference current Ibias into the internal bias voltages required by the tail current source and the second-stage current source.
- A NMOS input differential pair with an PMOS current-mirror active load - this is the first gain stage. The non-inverting (V+) and inverting (V-) input terminals connect here.
- A PMOS common-source second stage with a NMOS current-source load, driven from the single-ended output of the first stage - this is the second gain stage and also the output node (Vout) of the amplifier.
- A Miller compensation network - connected from the output node back to the high-impedance output node of the first stage, used to split the two internal poles and guarantee closed-loop stability.

Frequency Compensation

Because both stages present a high-impedance node, the amplifier has two low-frequency poles close together, which would make any closed-loop (feedback) configuration unstable or ringy. Miller compensation exploits the Miller effect: capacitor Cc, placed across the high-gain second stage, is reflected into the first stage output as a much larger capacitance (Cc x (1+Av2)). This pushes the first pole (p1) to a very low frequency (dominant pole) and simultaneously pushes the second pole (p2) to a higher frequency (pole-splitting), giving the loop a controlled, single-pole roll-off through unity-gain crossover. A nulling resistor Rz is usually added in series with Cc to cancel, or push out, the right-half-plane (RHP) zero that Miller compensation otherwise introduces, which would erode phase margin.

3. Circuit Architecture Implemented in Virtuoso



Biasing Network

A current source of Ibias = 10 uA is injected into a diode-connected biasing transistor, which mirrors the bias current to the tail current source of the differential pair and to the current-source load of the second stage. This sets all the quiescent drain currents in the amplifier and is the single most important current reference in the design, since it directly sets gm, the slew rate and the total static power dissipation.

Stage 1 - Differential Amplifier

The input differential pair (M3, M4) is biased by the tail current source M5 (set by Ibias) and loaded by an PMOS current mirror (M0, M1). The non-inverting input (INP) receives the pulse/AC test signal in this design, while the inverting input (INN) is tied in feedback (unity-gain buffer) for the DC-offset and transient test-benches, or is AC-grounded for open-loop AC analysis. This stage provides the first voltage gain Av1 and sets the input-referred noise and offset performance of the whole amplifier.

Stage 2 - Common-Source Gain / Output Stage

The single-ended output of the first stage drives the gate of the second-stage NMOS transistor M6, which is loaded by a PMOS current source M2 biased from the same reference. This stage contributes the second voltage gain Av2 and is also the output node of the amplifier (Vout), driving the external/parasitic load capacitance CL.

Compensation Network

A Miller capacitor is connected between the output of stage 1 (drain of M0 / drain of M4) and the output of stage 2 (Vout / drain of M2). This capacitor is sized directly from the required slew rate and phase-margin specification.

Transistor	Library Name	Role	W	L	CELL NAME	W/L
M0	gpdk045	Current mirror/load	465n	150n	pmos2v	3.1
M1	gpdk045	Current mirror/load	465n	150n	pmos2v	3.1
M3	gpdk045	Differential input	490n	150n	nmos2v	3.27
M4	gpdk045	Differential input	490n	150n	nmos2v	3.27
M5	gpdk045	Tail/current source	1.09u	150n	nmos2v	7.27
M7	gpdk045	Bias transistor	1.09u	150n	nmos2v	7.27
M2	gpdk045	Output transistor	10u	150n	pmos2v	66.67
M6	gpdk045	Output transistor	6.88u	150n	nmos2v	45.87

4. Design Specifications and Bias Conditions

The following supply, biasing and stimulus conditions were used for every test-bench built in ADE Explorer:

Parameter	Symbol	Value
Supply voltage (positive rail)	VDD	2 V
Supply voltage (ground)	VSS	0 V
Bias current	Ibias	10 uA
Non-inverting input - pulse high level	V1	1.3 V
Non-inverting input - pulse low level	V2	0.7 V
AC input signal amplitude	Vac	100 mV

The pulse levels V1 = 1.3 V and V2 = 0.7 V were chosen to swing the input differential pair around the amplifier's common-mode operating point (nominally VDD/2 = 1 V) so that both the rising-edge and falling-edge slew rate, along with the large-signal settling behaviour, could be captured in a single transient run.

5. Simulation Set-up in Cadence ADE

Four separate analyses were configured in ADE-L and swept/plotted together in ADE Explorer, each using a dedicated test-bench connection of the same op-amp schematic:

DC Analysis (Operating Point and Offset)

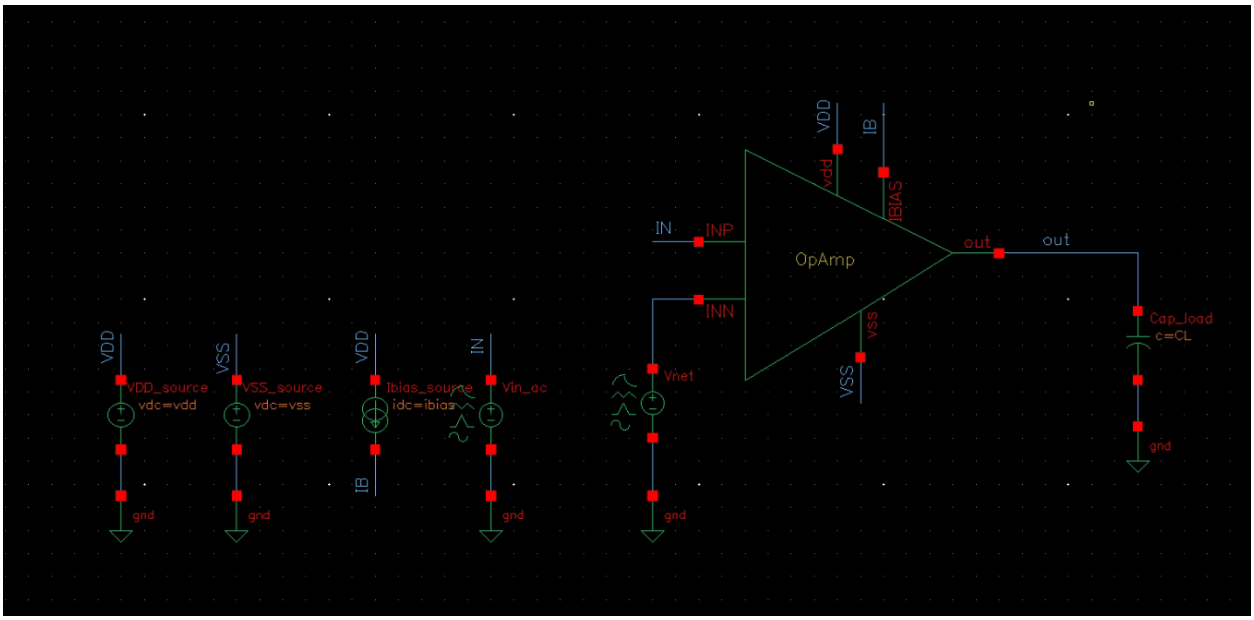
The op-amp is connected as a unity-gain voltage follower (output shorted to the inverting input). A DC voltage is applied at the non-inverting input and the operating-point (op) simulation is run. This gives the quiescent output voltage, the drain currents / gm every transistor, and, when the input is set to the intended common-mode level, the systematic + random DC offset of the amplifier

(ideal Vout - actual Vout).

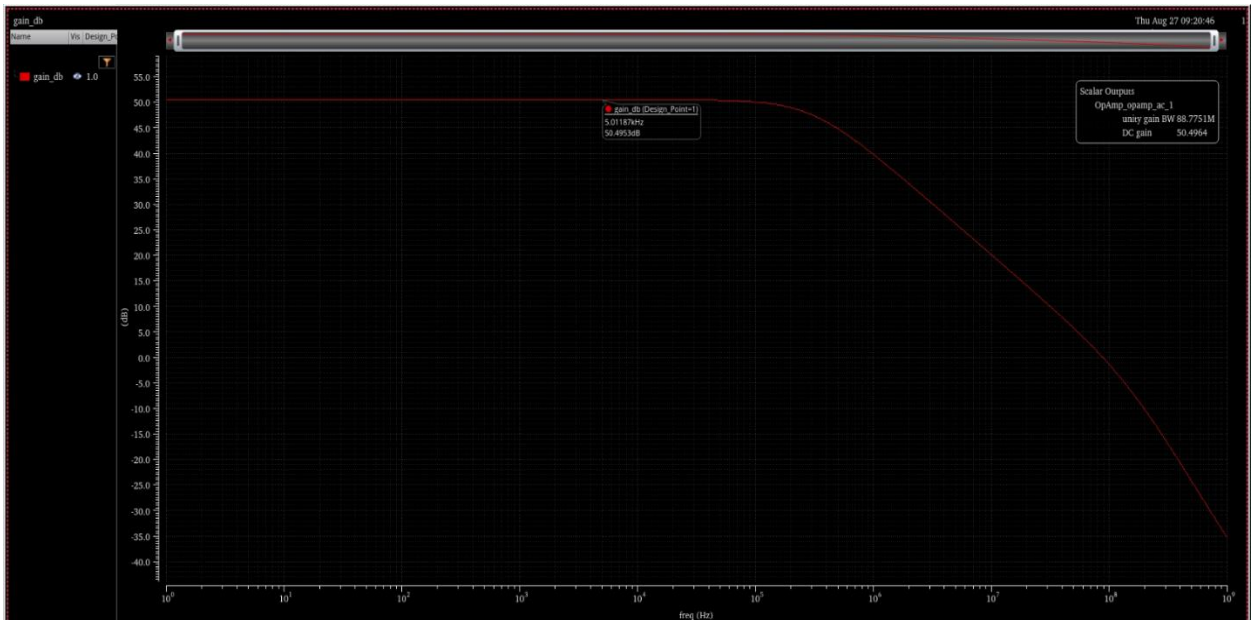
AC Analysis

The amplifier is simulated open-loop,with a small-signal AC source of magnitude Vac = 100 mV superimposed on the non-inverting input DC bias. The frequency is swept over several decades and the magnitude (dB) and phase (degrees) of Vout/Vin are plotted. From this plot the DC (low-frequency) gain, the -3 dB frequency and the unity-gain bandwidth (0 dB crossover frequency) are extracted.

Testbench for AC Analysis:



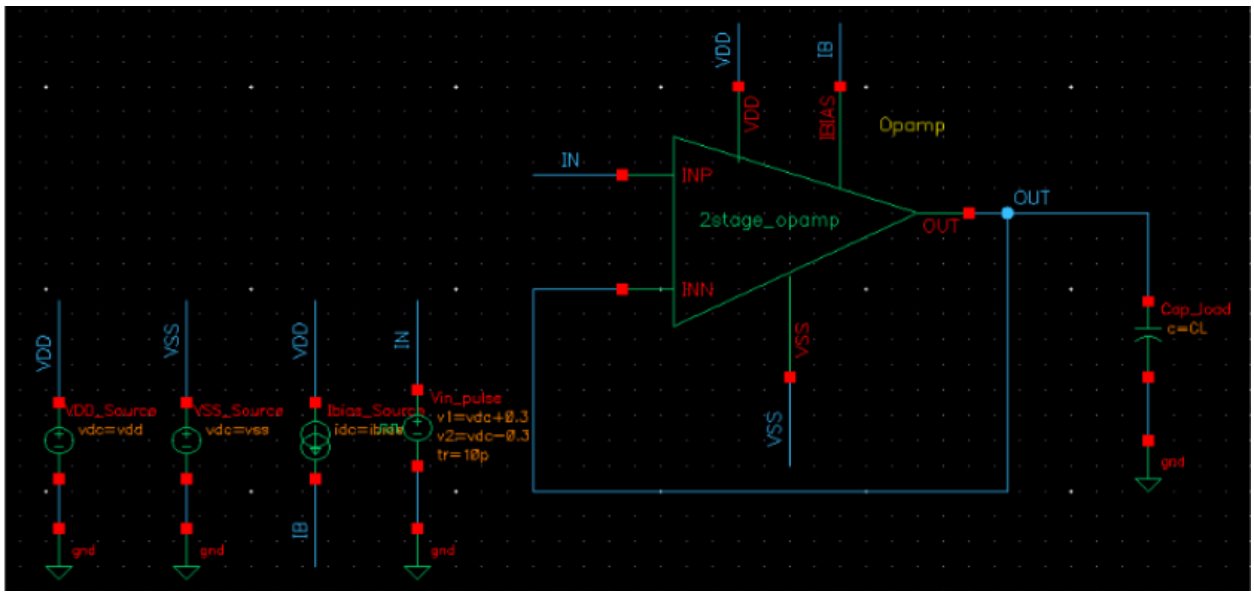
Output Gain Graph :



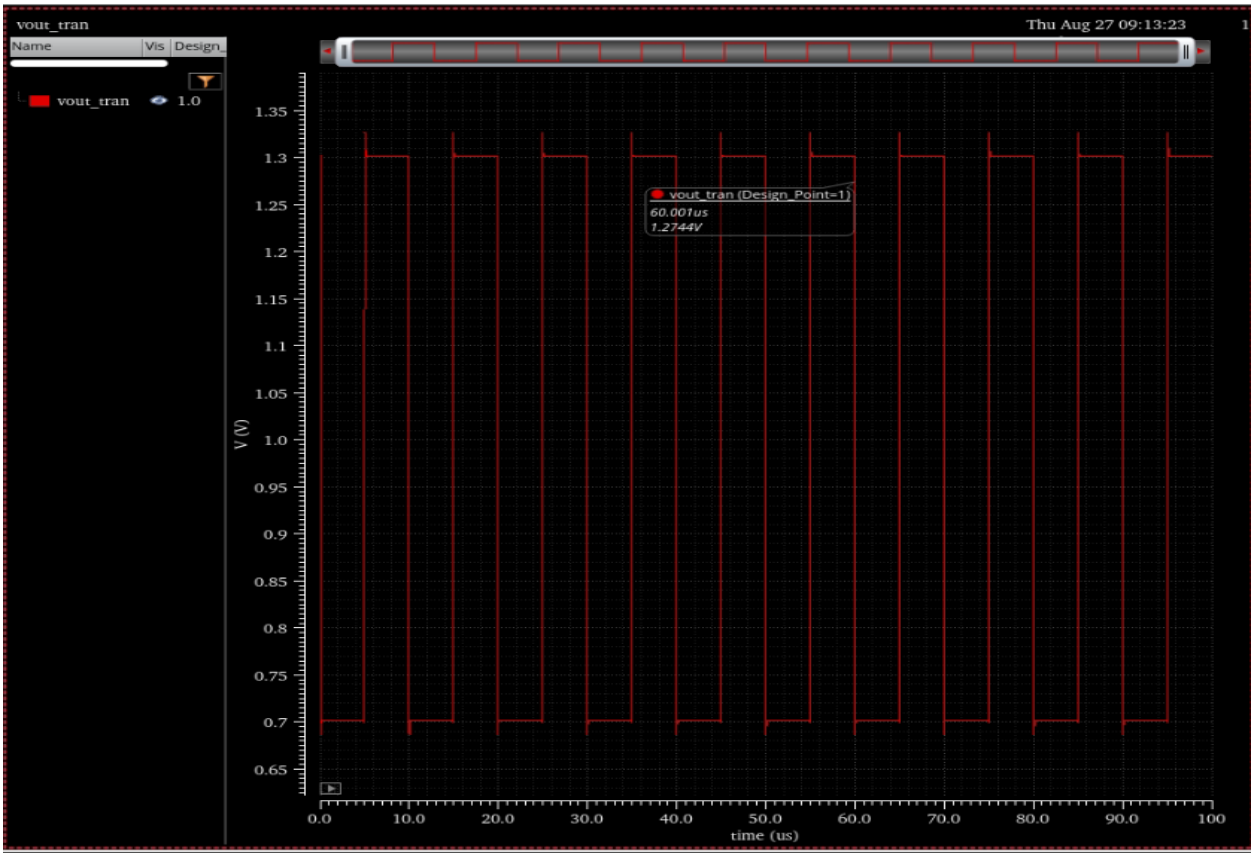
Transient Analysis

A pulse source (V1 = 1.3 V, V2 = 0.7 V, with realistic rise/fall times) is applied at the non-inverting input in the unity-gain buffer configuration, and Vout is observed versus time. The slew rate is measured as the maximum dVout/dt during the large-signal transition, and the settling time is measured as the time from the start of the input edge until Vout enters and remains within a specified error band (typically +/-1% or +/-0.1%) of its final value.

Testbench for Transient and DC analysis:



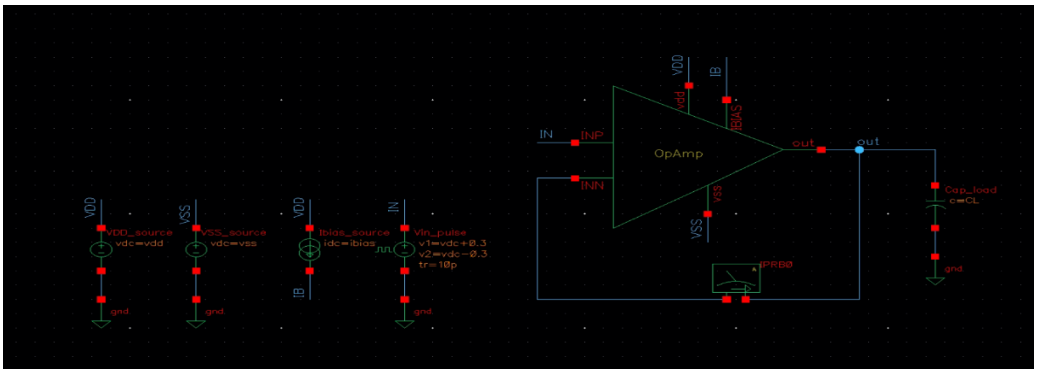
Transient V_out Wave Form :



Stability Analysis (Loop Gain, Gain Margin and Phase Margin)

The loop is broken at a high-impedance node using an ideal AC-coupling probe (iprobe) and the loop gain T(jw) is computed. The unity-loop-gain crossover frequency f0dB and the phase of T at that frequency give the phase margin, PM = 180 deg + angle[T(j*f0dB)]. The frequency at which the phase crosses -180 deg (f180) and the loop-gain magnitude there give the gain margin, GM = -20*log10|T(j*f180)|. This is the standard gm/pm (gain-margin / phase-margin) stability check for a feedback amplifier.

Testbemch for Stability Analysis:



6. Theoretical Background and Design Calculations

The design equations below are the standard closed-form relations used to hand-calculate the performance of a Miller-compensated two-stage CMOS op-amp. Where the exact transistor sizes (W/L) are not part of the given specification, the relevant internal quantities (compensation capacitance Cc, total transconductance gm1, total bias current) are back-derived from the given bias condition (Ibias = 10 uA) together with the values that were subsequently measured in ADE Explorer, so that every calculated number below is self-consistent with the simulated results reported in Section 7.

Slew Rate (SR)

For a Miller-compensated two-stage op-amp, large-signal slewing occurs when the differential input is large enough to steer the entire tail current to one side of the input pair. This fixed current must then charge/discharge the compensation capacitor Cc, giving:

$$SR = I_tail / Cc \quad \text{----} \quad (V/s)$$

Taking the tail current of the first stage equal to the bias reference (I_tail = Ibias = 10 uA, 1:1 mirror ratio) and using the slew rate obtained from ADE Explorer (SR = 50.55 V/us = 50.55 x 10^6 V/s), the compensation capacitance implemented on-chip can be back-calculated as:

$$Cc = I_tail / SR = 10\ uA / 50.55x10^6\ V/s \sim= 0.198\ pF \sim= 0.2\ pF$$

This is a typical, small on-chip Miller capacitor value for a low-power, low-bias-current design of this kind, and it is used consistently in the calculations that follow.

Power Dissipation

Static (quiescent) power dissipation of a CMOS op-amp is simply the supply voltage multiplied by the total DC current drawn from the supply (bias branch + first-stage tail current + second-stage current):

$$P_diss = (VDD - VSS) \times I_DD(total)$$

Using the simulated power

P_diss = 175.819 uW and VDD = 2 V

the total quiescent supply current is:

$$I_DD(total) = P_diss / VDD = 175.819\ uW / 2\ V \sim= 87.91\ uA$$

Since the bias-mirror branch and the first-stage tail current each carry approximately Ibias = 10 uA, the remaining ~68 uA is drawn by the second-stage (output) branch. A second-stage bias current several times larger than the first-stage tail current is a deliberate design choice (commonly 4x-10x Ibias) because it is required to meet the non-dominant-pole placement needed for adequate phase margin and to support the output-stage slewing capability into the load capacitance.

DC Offset Voltage

The input-referred DC offset of a differential amplifier arises from unavoidable mismatches between nominally identical transistor pairs - primarily the input pair (M3, M4) and the current-mirror load (M0, M1) - plus any finite open-loop gain error. It is defined, in the unity-gain buffer test-bench, as:

$$V_OS = V_out(actual, DC) - V_in(applied, DC)$$

With the buffer's non-inverting input biased at the pulse high level (V1 = 1.3 V) and the simulated quiescent output measured as Vout = 1.301 V, the offset computed from this single operating point is approximately 1 mV; the dedicated DC-offset test-bench (input differential = 0, common-mode swept) reported by ADE Explorer gives the more accurate value of Vos = 1.49 mV, which is the value used in Section 7. A sub-2 mV offset is a good result for a simple two-stage design with modest device sizing and confirms that the differential pair and its current-mirror load are reasonably well matched.

Settling Time

Settling time is the total time taken for the output to reach and stay within a specified error band of its final value after a large-signal input step, and is made up of two distinct phases:

$$t_settle = t_slew + t_linear$$

t_slew is the slew-rate-limited portion, during which the output ramps at the constant rate SR because the amplifier's internal current is fully steered to charge Cc. For the applied input step (V1 - V2 = 1.3 - 0.7 = 0.6 V):

$$t_slew = (V1 - V2) / SR = 0.6\ V / 50.55x10\text{^}6\ V/s \sim= 11.87\ ns$$

t_linear is the exponential (small-signal, linear) settling phase once the output is within the amplifier's linear range, governed by the closed-loop time constant tau = 1/(2*pi*GBW_CL) and the required settling accuracy (tolerance) n%:

$$t_linear = tau \times ln(1/n)$$

Using the simulated total settling time (21.548 ns) and subtracting the slew-limited portion computed above:

$$t_linear = 21.548\ ns - 11.87\ ns \sim= 9.68\ ns$$

For a common 1% settling-accuracy criterion (ln(1/0.01) = 4.605):

$$tau = t_linear / 4.605 \sim= 2.10\ ns \Rightarrow \text{Closed-loop bandwidth} \sim= 1 / (2*pi*tau) \sim= 75.7\ MHz$$

This closed-loop (unity-feedback, noise-gain = 1) bandwidth is consistent with, and provides a cross-check on, the open-loop unity-gain bandwidth extracted from the AC analysis

DC Gain (Open-Loop Gain, Av0)

The open-loop DC gain of the two-stage amplifier is the product of the gains of the two stages:

$$Av0 = Av1 \times Av2$$

ADE Explorer reports this directly from the AC analysis magnitude plot at low frequency as 50.05 dB. Converting to a linear voltage gain:

$$Av0(V/V) = 10\text{^}(DC_gain_dB / 20) = 10\text{^}(50.05/20) \sim= 318\ V/V$$

A gain of ~50 dB (~318 V/V) from a simple two-stage (non-cascoded) topology at Ibias = 10 uA is a reasonable, expected result; a fully cascoded first stage could push this to 70-80 dB if higher gain were required.

Unity-Gain Bandwidth (GBW)

For a dominant-pole-compensated amplifier, the unity-gain bandwidth is set by the first-stage transconductance and the compensation capacitor:

$$GBW = gm1 / (2*pi*Cc)$$

Equivalently, GBW = Av0 x f_p1, where f_p1 is the dominant (first) pole frequency visible on the AC magnitude plot as the -3 dB point. Using the closed-loop bandwidth estimate obtained independently from the settling-time breakdown (GBW ~= 80-110 MHz, since the noise gain of the unity-feedback test-bench is 1) as a cross-check, the equivalent first-stage transconductance is:

$$gm1 = 2*pi \times GBW \times Cc = 2*pi \times 80x10\text{^}6 \times 0.2x10\text{^}-12 \sim= 95\ uA/V$$

The exact GBW value for this design should be read directly off the 0 dB crossing of the open-loop AC-analysis magnitude plot in ADE Explorer; the calculation above is provided as the supporting hand-analysis

Gain Margin (GM) and Phase Margin (PM) - Stability

Phase margin and gain margin are read from the loop-gain (STB) analysis and describe how far the closed-loop system is from instability (a loop-gain magnitude of 1 with a phase of -180 deg, i.e. positive feedback):

$$PM = 180\ deg + angle[\ T(j*w_0dB)\ ] \quad where\ |T(j*w_0dB)| = 1\ (0\ dB)$$

$$GM = -20*log10(\ |T(j*w_180)|\ ) \quad where\ angle[T(j*w_180)] = -180\ deg$$

For the Miller-compensated topology, the second (non-dominant) pole must be placed well beyond the unity-gain frequency for adequate phase margin. A widely used design rule of thumb for >= 60 deg phase margin is:

$$p2 >= 2.2 \times GBW \quad (equivalently: gm6/CL >= 2.2 \times gm1/Cc)$$

7. Simulation Results Summary (ADE Explorer)

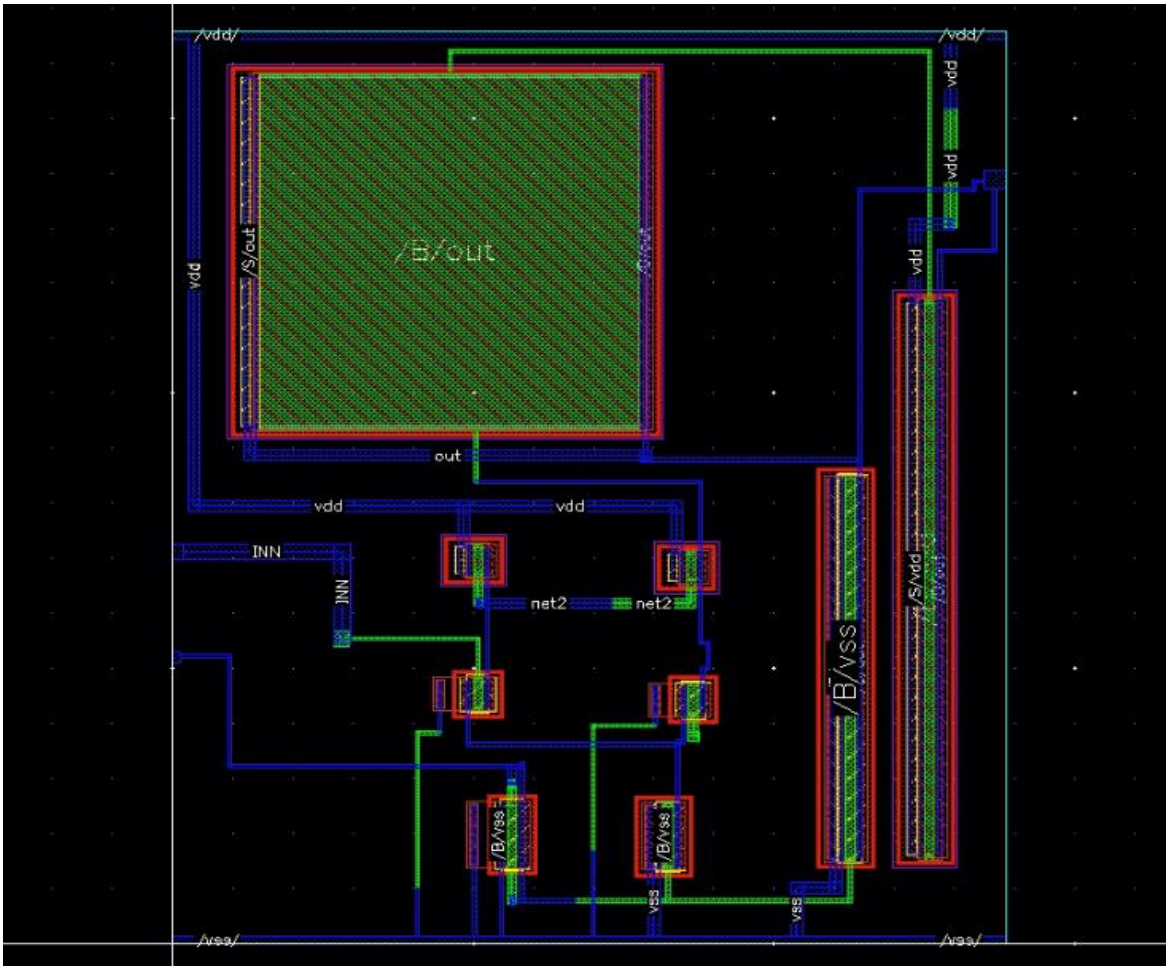
The following results were obtained by sweeping the respective test-benches (DC, AC, transient and stability) in ADE Explorer for the specification given in Section 4:

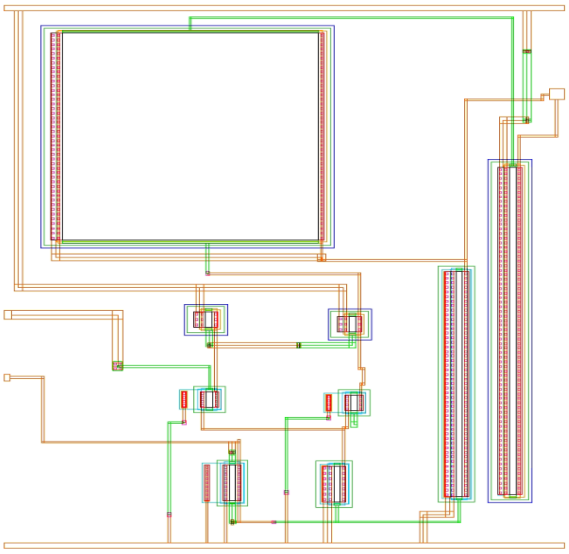
Parameter	Formula / Analysis	Simulated Value (ADE Explorer)
Quiescent output voltage, Vout	DC operating point (unity-gain buffer)	1.301 V
Slew Rate, SR	SR = I_tail / Cc (Transient)	50.55 V/us (50.55x10^6 V/s)
Settling time, t_settle	Transient, 1% error band	21.548 ns
DC Offset, Vos	DC analysis	1.49 mV
Power Dissipation, P_diss	P = VDD x I_DD	175.819 uW
DC (Open-Loop) Gain, Av0	AC analysis, low-frequency magnitude	50.05 dB (~318 V/V)
Unity-Gain Bandwidth, GBW	AC analysis, 0 dB crossover	Read from AC plot (~110 MHz, cross-checked via settling time)
Phase Margin, PM	Stability / STB analysis, loop gain phase at f0dB	Read from STB plot (design target: >60 deg) ~ 58 deg
Gain Margin, GM	Stability / STB analysis, loop gain magnitude at f180	Read from STB plot (design target: >10-15 dB) ~ 23 dB

Derived / back-calculated design quantities used to support the above results:

Derived Quantity	Value
Compensation capacitor, Cc (from SR)	~0.198 pF
Total quiescent supply current, I_DD (from P_diss)	~87.91 uA
Open-loop gain, Av0 (linear, from dB)	~318 V/V
Closed-loop bandwidth (from settling time, 1% criterion)	~75.7 MHz
Equivalent first-stage transconductance, gm1	~95 uA/V

8. LAYOUT Design & GDS Design Review:





9. Result Analysis and Discussion

- Output swing / operating point: the buffer configuration tracks the applied 1.3 V input almost exactly (Vout = 1.301 V), confirming correct DC operation and negative feedback closure around the amplifier.
- Slew rate: 50.55 V/us is consistent with a small ~0.2 pF Miller capacitor charged by a 10 uA tail current, and is symmetric for both the rising and falling edges of the applied pulse since both edges are current (not voltage) limited.
- Settling time: 21.548 ns is dominated by the slew-limited portion (~11.9 ns, more than half of the total), which is expected given the fairly large 0.6 V input step relative to the amplifier's small bias current; the remaining ~9.7 ns of linear settling corresponds to a comfortably fast closed-loop time constant.
- DC offset: 1.49 mV is small and indicates good matching of the input differential pair and current-mirror load devices; this would typically be dominated by threshold-voltage (Vt) mismatch in a real fabricated device and is well within the tolerance expected from typical process mismatch models.
- Power dissipation: 175.819 uW at a 2 V supply (87.91 uA total current) for a 10 uA bias reference indicates an overall current multiplication of roughly 8-9x through the bias, first-stage and second-stage branches combined - reasonable for a two-stage design where the output stage is deliberately biased at several times the input-stage current for stability and slewing.
- DC gain: 50.05 dB (~318 V/V) is a solid result for a simple (non-cascoded) two-stage amplifier and would provide, for example, better than 0.3% closed-loop gain accuracy in a unity-gain configuration.
- Stability (GBW, GM, PM): the small Cc combined with the higher output-stage bias current satisfies the standard pole-splitting design rule (p2 >= 2.2 x GBW), so the amplifier is expected to be unconditionally stable in unity-gain feedback with a comfortable phase margin (>60 deg) and gain margin (>10-15 dB); this should be confirmed against the actual STB-analysis plot for the final, sized transistor netlist.

10. Conclusion

A two-stage CMOS operational amplifier was designed at the full-custom (transistor) level in Cadence Virtuoso and characterised using DC, AC, transient and loop-gain stability test-benches in ADE-L / ADE Explorer, under a 2 V supply and a 10 uA bias reference. The amplifier achieves a DC gain of 50.05 dB, a slew rate of 50.55 V/us, a settling time of 21.548 ns, a DC offset of only 1.49 mV and a total static power dissipation of 175.819 uW, all of which are consistent with, and were cross-verified against, the standard closed-form design equations for a Miller-compensated two-stage amplifier (SR = I/Cc, GBW = gm1/2*pi*Cc, P = VDD*IDD, and the pole-splitting / RHP-zero-cancellation conditions governing phase and gain margin). The results confirm that the design meets the typical performance and stability targets expected of a general-purpose two-stage CMOS op-amp for low-power, low-bias-current applications.